

SKILL 5

To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output.

To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

Code:

```
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    printf("Enter command: ");
    fgets(input, sizeof(input), stdin);

    char *token = strtok(input, " \t\n");

    printf("Tokens:\n");

    while (token != NULL) {
        printf("%s\n", token);
        token = strtok(NULL, " \t\n");
    }
    return 0;
}
```

Output:

```
shashank@LAPTOP-24KPR0R8:~$ gcc token.c -o token
shashank@LAPTOP-24KPR0R8:~$ ./token
Enter command: ls
Tokens:
ls
```

b)

```
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    printf("Enter command: ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    if (strlen(input) == 0) {
        printf("Error: Empty command\n");
        return 0;
    }

    if (strcmp(input, "ls") == 0) {
        printf("Valid command: ls\n");
    }
    else if (strcmp(input, "pwd") == 0) {
        printf("Valid command: pwd\n");
    }
    else if (strcmp(input, "exit") == 0) {
        printf("Valid command: exit\n");
    }
    else {
        printf("Error: Invalid command\n");
    }
    return 0;
}
```

Output:

```
shashank@LAPTOP-24KPR0R8:~$ gcc parser.c -o parser
shashank@LAPTOP-24KPR0R8:~$ ./parser
Enter command: ls
Valid command: ls
```

